Solution The vertices are $A(-1,2)$, $B(k,-2)$ and $C(7,4)$

Area of triangle ABC is 22 sq.units

$$\frac{1}{2}\left\{(x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3)\right\} = 22$$

$$\frac{1}{2}\left\{(2 + 4k + 14) - (2k - 14 - 4)\right\} = 22$$

$$2k + 34 = 44 \text{ gives } 2k = 10 \text{ so } k = 5$$

Example 5.4 If the points $P(-1,-4)$, $Q(b,c)$ and $R(5,-1)$ are collinear and if $2b + c = 4$, then find the values of b and c.

Solution Since the three points $P(-1,-4)$, $Q(b,c)$ and $R(5,-1)$ are collinear,

$$\text{Area of triangle } PQR = 0$$

$$\frac{1}{2}\left\{(x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3)\right\} = 0$$

$$\frac{1}{2}\left\{(-c - b - 20) - (-4b + 5c + 1)\right\} = 0$$

$$-c - b - 20 + 4b - 5c - 1 = 0$$

$$b - 2c = 7 \quad \ldots(1)$$

Also, $2b + c = 4 \quad \ldots(2)$ (from given information)

Solving (1) and (2) we get $b = 3$, $c = -2$

Example 5.5 The floor of a hall is covered with identical tiles which are in the shapes of triangles. One such triangle has the vertices at $(-3,2)$, $(-1,-1)$ and $(1,2)$. If the floor of the hall is completely covered by 110 tiles, find the area of the floor.

Solution Vertices of one triangular tile are at

$(-3,2)$, $(-1,-1)$ and $(1,2)$

Area of this tile $= \frac{1}{2}\left\{(3 - 2 + 2) - (-2 - 1 - 6)\right\}$ sq.units

$= \frac{1}{2}(12) = 6$ sq.units

Since the floor is covered by 110 triangle shaped identical tiles,

Area of floor $= 110 \times 6 = 660$ sq.units

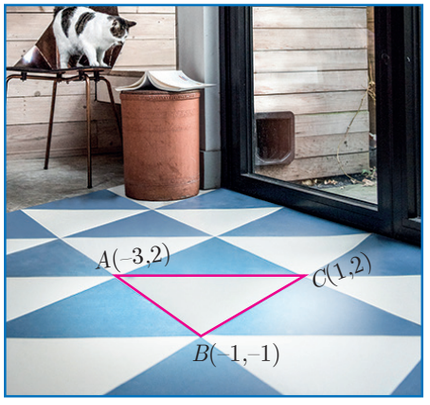


Fig. 5.11

Example 5.6 Find the area of the quadrilateral formed by the points $(8,6)$, $(5,11)$, $(-5,12)$ and $(-4,3)$.

Solution Before determining the area of quadrilateral, plot the vertices in a graph.

Let the vertices be $A(8,6)$, $B(5,11)$, $C(-5,12)$ and $D(-4,3)$

Therefore, area of the quadrilateral $ABCD$

10th_Maths_Chapter 5_English.indd 209 12/27/2024 3:16:07 PM